Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2025-26

Class:	BE	Semester:	VIII
Course Code:		Course Name:	RS

Name of Student:	BARI ANKIT VINOD
Roll No. :	61
Assignment No.:	1
Title of Assignment:	
Date of Submission:	
Date of Correction:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Completeness	5	5
Demonstrated Knowledge	3	2
Legibility	2	2
Total	10	9

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Completeness	5	3-4	1-2
Demonstrated Knowledge	3	2	1
Legibility	2	1	0

Checked by

Name of Faculty : Mr. Raunak Joshi

Signature :

Date :

Q. 1) Given two matrices A and B, Calculate A+B & 2A-3B

→ given

$$A = \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix} , \quad D = \begin{bmatrix} 5 & -1 \\ 0 & 2 \end{bmatrix}$$

(a) $C \times D$

$$\therefore C \times D = \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 5 & -1 \\ 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 15 & -1 \\ 10 & 6 \end{bmatrix}$$

(d) given,

$$C = \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix} , \quad D = \begin{bmatrix} 5 & -1 \\ 0 & 2 \end{bmatrix}$$

$D \times C$

$$\therefore D \times C = \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 5 & -1 \\ 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 13 & 1 \\ 4 & 8 \end{bmatrix}$$

Hence,

$$C \times D = \begin{bmatrix} 15 & -1 \\ 10 & 6 \end{bmatrix} \quad \& \quad D \times C = \begin{bmatrix} 13 & 1 \\ 4 & 8 \end{bmatrix}$$

$$\therefore C \times D \neq D \times C$$

Q. 2) Consider the following matrix. compute the product $C \times D$ determine if matrix multiplication is commutative by comparing $C \times D$ & $D \times C$.

→ given,

$$A = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix} \qquad B = \begin{bmatrix} 4 & -2 \\ 0 & 5 \end{bmatrix}$$

(a) $A + B =$

$$\therefore A + B = \begin{bmatrix} 2+4 & 1+(-2) \\ -1+0 & 3+5 \end{bmatrix} = \begin{bmatrix} 6 & -1 \\ -1 & 8 \end{bmatrix}$$

(b) given,

$$A = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix} \qquad B = \begin{bmatrix} 4 & -2 \\ 0 & 5 \end{bmatrix}$$

$$\therefore 2A = \begin{bmatrix} 4 & 2 \\ -2 & 6 \end{bmatrix}$$

&

$$3B = \begin{bmatrix} 12 & -6 \\ 0 & 15 \end{bmatrix}$$

$$\therefore 2A - 3B = \begin{bmatrix} 4-12 & 2-(-6) \\ -2-0 & 6-15 \end{bmatrix}$$

$$= \begin{bmatrix} -8 & 8 \\ -2 & -9 \end{bmatrix}$$